

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 05 · Arithmetic Sequences

With Answers

Questions with full worked solutions

- 16 pages
- Past papers up to January 2026

PURE 2 | TOPIC 05

Arithmetic Sequences

Your classified route

6

QUESTIONS

41

MARKS

2

FAMILIES

3

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 05 | FAMILY 01

Unknown Terms

WHAT THIS FAMILY ASKS

Recognise the unknown terms demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 Use simultaneous equations when two term positions are given.

2 classified questions

14 total marks

T05

05_SOLVE

05_SOLVE_HV02

CHAPTER
Arithmetic Sequences

FAMILY
Unknown terms

VARIANT
Use simultaneous equations
when two term positions are
given.

7 marks

WMA12/01 OCTOBER 2023

Question 8

Past-paper question

WMA12-01_2023_Oct_Q08

8. In a large theatre there are n rows of seats, where n is a constant.

The number of seats in the first row is a , where a is a constant.

In each subsequent row there are 4 more seats than in the previous row so that

- in the 2nd row there are $(a + 4)$ seats
- in the 3rd row there are $(a + 8)$ seats
- the number of seats in each row form an **arithmetic** sequence

Given that the **total** number of seats in the first 10 rows is 360

- (a) find the value of a .

(2)

Given also that the total number of seats in the n rows is 2146

- (b) show that

$$n^2 + 8n - 1073 = 0$$

(2)

- (c) Hence

- state the number of rows of seats in the theatre,
- find the maximum number of seats in any one row.

(3)

T05 Arithmetic Sequences

05_SOLVE Unknown terms

05_SOLVE_HV02 Use simultaneous equations when two term positions are given.

WORKED SOLUTION

Question 8

WMA12/01 October 2023

WMA12-01_2023_Oct_Q08

1

Use the arithmetic sum formulaThe first term is a and the common difference is 4.

2

Use the first 10 rows

$$S_{10} = \frac{10}{2} (2a + 9 \cdot 4) = 360.$$

3

Find a

$$5(2a + 36) = 360, \text{ so } 10a + 180 = 360 \text{ and } a = 18.$$

4

Write S_n

$$S_n = \frac{n}{2} (2(18) + (n-1)4) = \frac{n}{2} (4n + 32) = 2n^2 + 16n.$$

5

Use the total number of seats

$$\text{Since the total is 2146, } 2n^2 + 16n = 2146.$$

6

Show the equation

$$\text{Dividing by 2 and rearranging gives } n^2 + 8n - 1073 = 0.$$

7

Solve for n

$$n = \frac{-8 \pm \sqrt{8^2 + 4(1073)}}{2} = \frac{-8 \pm 66}{2}, \text{ so } n = 29 \text{ or } n = -37.$$

8

Choose the row count

The number of rows must be positive, so there are 29 rows.

9

Find the maximum row size

$$\text{The largest row is the last row: } 18 + (29 - 1)4 = 18 + 112 = 130 \text{ seats.}$$

FINAL ANSWER

WMA12-01_2023_Oct_Q08

(a) $a = 18$. (b) **Proven.** (c)(i) **29 rows.** (c)(ii) **130 seats.**

T05

05_SOLVE

05_SOLVE_HV02

CHAPTER
Arithmetic Sequences

FAMILY
Unknown terms

VARIANT
Use simultaneous equations
when two term positions are
given.

7 marks

WMA12/01 MAY/JUNE 2024

Question 2

Past-paper question

WMA12-01_2024_MayJune_Q02

2. **In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

In an arithmetic series,

- the sixth term is 2
- the sum of the first ten terms is -80

For this series,

- (a) find the value of the first term and the value of the common difference.

(4)

- (b) Hence find the smallest value of n for which

$$S_n > 8000$$

(3)

T05 Arithmetic Sequences

05_SOLVE Unknown terms

05_SOLVE_HV02 Use simultaneous equations when two term positions are given.

WORKED SOLUTION

Question 2

WMA12/01 May/June 2024

WMA12-01_2024_MayJune_Q02

1

Use the sixth termFor an arithmetic series, $u_6 = a + 5d = 2$.

2

Use the sum of ten terms $S_{10} = \frac{10}{2}(2a + 9d) = -80$, so $2a + 9d = -16$.

3

Solve simultaneouslyFrom $a + 5d = 2$, $a = 2 - 5d$. Substitute into $2a + 9d = -16$.

4

Find a and d $2(2 - 5d) + 9d = -16$, so $4 - d = -16$ and $d = 20$. Then $a = 2 - 5(20) = -98$.

5

Write S_n $S_n = \frac{n}{2}(2a + (n-1)d) = \frac{n}{2}(-196 + 20n - 20) = 10n^2 - 108n$.

6

Set up the inequalityWe need $10n^2 - 108n > 8000$.

7

Solve the boundary $10n^2 - 108n - 8000 = 0$ gives $5n^2 - 54n - 4000 = 0$.

8

Find the positive threshold $n = \frac{54 + \sqrt{82916}}{10} = 34.195 \dots$

9

Choose the smallest integerTherefore the smallest integer value with $S_n > 8000$ is $n = 35$.**FINAL ANSWER**

WMA12-01_2024_MayJune_Q02

(a) First term = -98 , common difference = 20 . (b) $n = 35$.

TOPIC 05 | FAMILY 02

Series Sum

WHAT THIS FAMILY ASKS

Recognise the series sum demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

V02 If last term is unknown, use $l = a + (n-1)d$ first.

4 classified questions

27 total marks

T05

05_SUM

05_SUM_CORE

CHAPTER

Arithmetic Sequences

FAMILY

Series sum

VARIANT

core family move

13 marks

WMA12/01 MAY/JUNE (R) 2024

Question 8

Past-paper question

WMA12-01_2024_MayJuneR_Q08

8. (i) (a) In an **arithmetic** series the first term is a and the common difference is d .

Show that

$$S_n = \frac{n}{2} \{2a + (n-1)d\} \quad (3)$$

- (b) Hence find

$$900 + 892 + 884 + \dots + 500 \quad (3)$$

- (ii) Given that the first three terms of a **geometric** series are

$$k+4 \quad k-2 \quad 11-k$$

where k is a constant,

- (a) show that

$$2k^2 - 11k - 40 = 0 \quad (3)$$

Given also that this series is convergent,

- (b) find the value of S_∞ (4)

T05 Arithmetic Sequences

05_SUM Series sum

05_SUM_CORE core family
move

WORKED SOLUTION

Question 8

WMA12/01 May/June (R) 2024

WMA12-01_2024_MayJuneR_Q08

1

Write the first and last termsThe first term is a and the n th term is $a + (n - 1)d$.

2

Use the arithmetic sum formula $S_n = \frac{n}{2}(\text{first term} + \text{last term})$.

3

Show the formulaTherefore $S_n = \frac{n}{2}\{a + [a + (n - 1)d]\} = \frac{n}{2}\{2a + (n - 1)d\}$.

4

Identify the arithmetic seriesFor $900 + 892 + 884 + \dots + 500$, the first term is 900, the common difference is -8 , and the last term is 500.

5

Find the number of terms $500 = 900 + (n - 1)(-8)$, so $n - 1 = 50$ and $n = 51$.

6

Find the sum $S = \frac{51}{2}(900 + 500) = 35700$.

7

Use the geometric conditionFor $k + 4$, $k - 2$, $11 - k$ to be consecutive geometric terms, $(k - 2)^2 = (k + 4)(11 - k)$.

8

Show the quadratic $k^2 - 4k + 4 = -k^2 + 7k + 44$, so $2k^2 - 11k - 40 = 0$.

9

Solve for k $2k^2 - 11k - 40 = 0$ gives $k = 8$ or $k = -\frac{5}{2}$.

10

Use convergenceIf $k = 8$, the terms are 12, 6, 3 and the common ratio is $\frac{1}{2}$, so the series is convergent. If $k = -\frac{5}{2}$, the common ratio is -3 , so it is not convergent.

11

Find S infinity

Use $k = 8$: $S_{\infty} = \frac{12}{1-1/2} = 24$.

FINAL ANSWER

WMA12-01_2024_MayJuneR_Q08

(i)(a) **Proven.** (i)(b) 35700. (ii)(a) **Proven.** (ii)(b) $S_{\infty} = 24$.

DR ESLAM AHMED

T05

05_SUM

05_SUM_CORE

CHAPTER
Arithmetic Sequences

FAMILY
Series sum

VARIANT
core family move

4 marks

WMA12/01 JANUARY 2025
Question 1

Past-paper question

WMA12-01_2025_Jan_Q01

1. The arithmetic series S is given by

$$S = 2 + 5 + 8 + 11 + \dots + 254$$

Find

- (a) the number of terms in the series,

(2)

- (b) the sum of the series.

(2)

T05 Arithmetic Sequences

05_SUM Series sum

05_SUM_CORE core family
move

WORKED SOLUTION

Question 1

WMA12/01 January 2025

WMA12-01_2025_Jan_Q01

1

Identify the arithmetic series

The first term is 2, the common difference is 3, and the last term is 254.

2

Find the number of terms

Use $254 = 2 + (n - 1)3$. Then $252 = 3(n - 1)$, so $n - 1 = 84$ and $n = 85$.

3

Use the sum formula

$S_n = \frac{n}{2}(a + l)$, where a is the first term and l is the last term.

4

Find the sum

$$S = \frac{85}{2}(2 + 254) = 85 \times 128 = 10880.$$

FINAL ANSWER

(a) 85. (b) 10880.

WMA12-01_2025_Jan_Q01

T05

05_SUM

05_SUM_CORE

CHAPTER

Arithmetic Sequences

FAMILY

Series sum

VARIANT

core family move

4 marks

WMA12/01 OCTOBER 2025

Question 1

Past-paper question

WMA12-01_2025_Oct_Q01

1. An arithmetic series starts

$$78 + 75 + 72 + \dots$$

Find

- (a) the 18th term in the series,

(2)

- (b) the sum of the first 100 terms of the series.

(2)

T05 Arithmetic Sequences

05_SUM Series sum

05_SUM_CORE core family
move

WORKED SOLUTION

Question 1

WMA12/01 October 2025

WMA12-01_2025_Oct_Q01

1

Identify the sequence

The first term is 78 and the common difference is $75 - 78 = -3$.

2

Find the eighteenth term

$$u_{18} = 78 + 17(-3) = 27.$$

3

Use the sum formula

$$S_{100} = \frac{100}{2}\{2(78) + 99(-3)\}.$$

4

Evaluate the sum

$$S_{100} = 50(156 - 297) = 50(-141) = -7050.$$

FINAL ANSWER

(a) 27. (b) -7050.

WMA12-01_2025_Oct_Q01

T05

05_SUM

05_SUM_HV01

CHAPTER

Arithmetic Sequences

FAMILY

Series sum

VARIANT

If last term is unknown, use
 $(l=a+(n-1)d)$ first.

6 marks

WMA12/01 OCTOBER 2021

Question 5

Past-paper question

WMA12-01_2021_Oct_Q05

5. A company that owned a silver mine

- extracted 480 tonnes of silver from the mine in year 1
- extracted 465 tonnes of silver from the mine in year 2
- extracted 450 tonnes of silver from the mine in year 3

and so on, forming an arithmetic sequence.

(a) Find the mass of silver extracted in year 14

(2)

After a total of 7770 tonnes of silver was extracted, the company stopped mining.

Given that this occurred at the end of year N ,

(b) show that

$$N^2 - 65N + 1036 = 0$$

(3)

(c) Hence, state the value of N .

(1)

T05 Arithmetic Sequences

05_SUM Series sum

05_SUM_HV01 If last term is unknown, use $l = a + (n-1)d$ first.

WORKED SOLUTION

Question 5

WMA12/01 October 2021

WMA12-01_2021_Oct_Q05

1

Identify the arithmetic sequence

The first term is 480 and the common difference is $465 - 480 = -15$.

2

Find year fourteen

$u_{14} = 480 + 13(-15) = 285$, so 285 tonnes were extracted in year 14.

3

Set up the total

The total after N years is $S_N = \frac{N}{2}\{2(480) + (N-1)(-15)\}$.

4

Use the total extraction

$\frac{N}{2}(975 - 15N) = 7770$.

5

Show the quadratic

Multiplying by 2 and rearranging gives $15N^2 - 975N + 15540 = 0$, so $N^2 - 65N + 1036 = 0$.

6

Solve and choose N

$N = 28$ or 37 . Since extraction amounts would become negative before year 37, the stopping year is $N = 28$.

FINAL ANSWER

WMA12-01_2021_Oct_Q05

(a) 285 tonnes. (b) Proven: $N^2 - 65N + 1036 = 0$. (c) $N = 28$.